

National University of Computer and Emerging Sciences
Lahore Campus

Computer Networks (CS3001)

Date: April 6th 2026

Course Instructor(s)

Ms. Sarah Khaleel, Mr. Nauman Moazzam Hayat

Sessional-II Exam

Total Time: 1 Hours

Total Marks: 35

Total Questions: 5

Roll No

Section

Student Signature

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- Instruction/Notes:**
- Attempt all questions (except Q#1 & Q#5) on the separately provided answer sheet.
 - Q#1 and Q#5 are required to be answered on this question paper as per instructions provided in the question statement.
 - Ensure that question paper is attached to the provided answer sheet.
 - In case you have used rough sheets, they should **NOT** be attached to the answer sheet.
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CLO 1 (Q#1): Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

Q1: Answer the following MCQs by **encircling** the correct option on the question paper: Any erasing, cutting and overwriting is not allowed. (Answers given in the answer sheet will **not be marked**.) [1 * 5 = 5 Marks]

1. A TCP Tahoe sender has: cwnd = 12 MSS, ssthresh = 16 MSS. A packet loss is detected via 3 duplicate ACKs. What will be the new cwnd immediately after this event?

- A. 6 MSS B. 1 MSS C. 8 MSS D. 11 MSS

Answer: B

2. A Go-Back-N protocol uses: Window size = 4, Sender transmits packets: 0,1,2,3. Packet 1 is lost. Packets 0, 2 and 3 arrive at receiver. How many packets will be retransmitted?

- A. 1 B. 2 C. 3 D. 4

Answer: C

3. A TCP sender has: cwnd = 14 MSS, rwnd = 10 MSS. What is the maximum number of unacknowledged segments that can be sent?

- A. 4 B. 10 C. 14 D. 24

Answer: B

4. The service model of the internet's network layer (IPv4) can be best described as:

- A. Unreliable B. Fast C. Unreliable & Fast D. Best Effort

Answer: D

5. A host obtains an IP address for itself via:

- A. DHCP B. DNS C. HTTP D. FTP

Answer: A

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CLO 2 (Q#2,3 and 4): Demonstrate the basics of network concepts using state-of-the-art network tools/techniques.

Q2: UDP, TCP & IP use 1s complement for their checksums. Suppose you have the following three 8-bit bytes: 01010011, 01100110, 01110100. What is the 1s complement of the sum of these 8-bit bytes? (Note that although UDP, TCP or IP use 16-bit words in computing the checksum, for this problem you are being asked to consider 8-bit sums.) **[5 Marks]**

Solution: Note, wrap around if overflow.

$$\begin{array}{r} 01010011 \\ + 01100110 \\ \hline 10111001 \end{array}$$

$$\begin{array}{r} 10111001 \\ + 01110100 \\ \hline 00101101 \end{array}$$

One's complement = 11010001.

Q3: A TCP sender uses the following parameters:

[2 * 3 = 6 Marks]

- Initial EstimatedRTT = 200 ms, DevRTT = 20 ms
- $\alpha = 0.125$, $\beta = 0.25$
- New SampleRTT = 260 ms

(a) Compute the new EstimatedRTT

Solution: $\text{DevRTT} = (1 - \beta) \cdot \text{DevRTT} + \beta \cdot |\text{SampleRTT} - \text{EstimatedRTT}|$
Substitute values:
 $= (1 - 0.25)(20) + 0.25 \cdot |260 - 200|$
 $= 0.75 \times 20 + 0.25 \times 60$
 $= 15 + 15$
 $= 30 \text{ ms}$

(b) Compute the new DevRTT

Solution: $\text{EstimatedRTT} = (1 - \alpha) \cdot \text{EstimatedRTT} + \alpha \cdot \text{SampleRTT}$
 $= 0.875 \times 200 + 0.125 \times 260$
 $= 175 + 32.5$
 $= 207.5 \text{ ms}$

(c) Compute the new TimeoutInterval

Solution: $\text{Timeout} = \text{EstimatedRTT} + 4 \cdot \text{DevRTT}$
 $= 207.5 + 4 \times 30$
 $= 207.5 + 120$
 $= 327.5 \text{ ms}$

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Q4(a): A cloud provider has four separate clusters in a data center. Find the most specific summary route (in dotted decimal CIDR notation including prefix) that includes all four of these subnets: **[5 Marks]**

- i) 172.16.12.0/24
- ii) 172.16.13.0/24
- iii) 172.16.14.0/24
- iv) 172.16.15.0/24

Solution:

Look at the third octet in binary:

172.16.12: 00001100

172.16.13: 00001101

172.16.14: 00001110

172.16.15: 00001111

The first 6 bits (000011) are common. 16 bits (first two octets) + 6 bits in the 3rd octet = 22

Summary Route = 172.16.12.0/22

Q4(b): A router receives a packet with the destination IP 172.16.14.200. The router's forwarding table is:

Prefix Entry	Output Interface
172.16.0.0/16	0
172.16.14.0/23	1
172.16.14.128/25	2
172.16.14.192/26	3

On which interface will the packet be forwarded to?

[4 Marks]

Solution:

Answer is Output **interface 3**.

The destination IP **172.16.14.200** matches **all four** entries in the table.

The reason is /26 is the longest prefix (most specific mask). Even though the packet "fits" in the /16, /23 and /25, the router always chooses the most specific (longest) match.

CLO 3 (Q#5): Demonstrate various classical routing and switching protocols via simulations.

Q5: Answer to this question is required to be provided **on the question paper** by completing the given table.

A student brings a brand-new laptop into a university lab. The laptop is configured for dynamic IP addressing. The lab's DHCP server has the IP address 10.1.1.254 and is sitting on the 10.1.1.0/24 network. Please fill in the below table with the missing entries for the DHCP messages exchange between the laptop & the DHCP Server in the **correct order**. **[0.5 * 20 = 10 Marks]**

Step	DHCP Message Name	Source IP Address	Source Port Number	Destination IP Address	Destination Port Number
1					
2					
3					
4					

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Solution:

Step	DHCP Message Name	Source IP Address	Source Port Number	Destination IP Address	Destination Port Number
1	DHCP Discover	0.0.0.0	68	255.255.255.255	67
2	DHCP Offer	10.1.1.254	67	255.255.255.255	68
3	DHCP Request	0.0.0.0	68	255.255.255.255	67
4	DHCP ACK	10.1.1.254	67	255.255.255.255	68